MILITARY SPECIFICATION

ROPE, FIBROUS, POLYESTER/POLYPROPYLENE DUAL FIBER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three strand, polyester/polypropylene dual fiber rope.

1.2 Classification. The polyester/polypropylene dual fiber rope shall be furnished in one type in the circumference (size) as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4020

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.2 Government or supplier purchases. The requirements specified in 3.9, 3.10 and 3.11 apply only to rope purchased directly by the Government. All other requirements apply to rope purchased as a component for an end item by a contractor and to rope purchased directly by the Government.

3.3 Standard sample. Dyed rope shall match the standard for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.4 Materials.

3.4.1 Polyester. The polyester portion of the rope shall be bright, virgin, continuous filament polyester fiber of suitable tenacity to meet the requirements of this specification.

3.4.2 Polypropylene. The polypropylene portion of the rope shall be virgin, continuous, monofilament fibers and of suitable strength to meet all the requirements of this specification; a minimum softening point of 300°F and a specific gravity no greater than 0.91.

3.4.2.1 Cover yarns. The polypropylene portion of these yarns shall be virgin, continuous, monofilament fiber ranging in size from 100 to 200 denier per filament and shall contain heat and ultraviolet light stabilizers.

3.4.2.2 Inside yarns. These yarns shall be virgin, continuous, monofilament or may be continuous slit or fibrillated film polypropylene, and shall contain heat but not necessarily ultraviolet light stabilizers.

3.5 Construction.

3.5.1 Lay. The rope shall be three strand polyester/polypropylene and the lay of the rope shall be right turn or "Z" lay.

3.5.2 Strands. The strands shall be constructed as indicated below.

3.5.2.1 Cover yarns. The cover yarns shall be a combination of polyester over polypropylene and shall contain a minimum of 50 percent by weight of polyester fiber when tested as specified in 4.4.5. For rope sizes from 5/8 to 2-1/4 inch circumference inclusive, each strand shall be made of either single ply or three ply yarns. For rope sizes over 2-1/2 inches circumference, each strand shall be made of three ply cover yarns.

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3.5.2.2 Inside yarns. In rope sizes of 5/8 to 2-1/4 inch circumference, if one single yarn is used inside, it shall be left (S) twist.

3.5.3 In the preparation of the constituent rope strands, the following shall apply:

(1) For ropes containing 3 ply cover yarns.

Component	Twist direction
Singles yarns	S
3 ply uarns	Z
Strands	S
Rope	Z

(2) For ropes containing singles cover yarns.

Component	Twist direction
Singles inside yarns	S
Singles cover yarns	Z
Strands	S
Rope	Z

3.6 Physical requirements. The rope shall be of three strands and shall conform to the requirements specified in table I when tested as specified in 4.4.3.

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TABLE I. Physical requirements

Circumference (Nominal)	Tolerance (±)	Diameter 1/ (approx)	Length per pound (min) 2/
in	in	in	ft
5/8	1/16	3/16	100.0
3/4	1/16	1/4	58.8
1	1/16	5/16	38.5
1-1/8	1/16	3/8	26.3
1-1/4	1/16	7/16	18.9
1-1/2	1/16	1/2	14.7
1-3/4	1/16	9/16	11.9
2	1/8	5/8	10.0
2-1/4	1/8	3/4	7.6
2-1/2	1/8	13/16	6.3
2-3/4	1/8	7/8	5.3
3	1/8	1	4.3
3-1/4	3/16	1-1/16	3.7
3-1/2	3/16	1-1/8	3.3
3-3/4	1/4	1-1/4	2.9

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TABLE I. Physical requirements (cont'd)

Circumference (Nominal)	Break force (min)	Length per reel (min)	Mass per reel (approx) 1/	Length per 30 Crowns (max)
in	lb	ft	lb	in
5/8	720	2250	24	8-1/8
3/4	1130	2250	41	9
1	1710	2250	63	12-5/16
1-1/8	2430	1600	65	13-7/8
1-1/4	3150	1200	68	15-3/16
1-1/2	3960	1200	87	17-3/4
1-3/4	4860	1200	107	20-3/16
2	5760	1200	127	22-9/16
2-1/4	7560	1200	167	25-1/16
2-1/2	9180	1200	204	27-3/8
2-3/4	10800	1200	241	29-11/16
3	13100	1200	297	31-15/16
3-1/4	15200	1200	346	34-5/8
3-1/2	17400	1200	386	37
3-3/4	19800	600	220	39-3/8

1/ Not a specification requirement - included for information purposes only.

2/ A minus tolerance of 10 percent is allowed for dyed rope.

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3.6.1 Elongation. The elongation of the rope shall not exceed 35 percent when tested as specified in 4.4.5.

3.7 Finish. No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall not exceed 4.0 percent when tested as specified in 4.4.5.

3.8 Color. Unless otherwise specified (see 6.2), the color of the rope shall be natural. When colored rope is specified, the rope shall be dyed to match an approved standard shade of Olive Drab 7 and both the polyester and polypropylene components of the cover yarns shall be dyed (see 6.3).

3.8.1 Colorfastness. Dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.4.5. When no standard sample has been established, dyed rope shall show good fastness to weathering when tested as specified in 4.4.5.

3.9 Identification. Each reel, spool of rope shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and have a minimum tearing resistance of both directions (total) of 850 grams when tested as specified in UU-T-81. The label shall be attached in such a manner as to remain in place and be clearly legible until all rope has been removed. All ticket or label entries shall be legibly printed, stamped or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock number
Color
Specification number
Length and circumference (size)
Contract number and date
Date of manufacture (Month and Year)
Supplier's name

3.9.1 Identification marker. The manufacturer shall identify his product by inserting a kraft paper or water repellent cotton marker within one strand in all ropes larger than 1-1/8 inch circumference. The marker shall be completely enveloped by the cover yarns in the strand in which it is inclosed. Unless otherwise specified, the manufacturer's name, year of manufacture and type of fiber (POLYESTER/POLYPROPYLENE) shall be clearly printed in block type on the marker. The printing shall not be affected upon exposure to water or mineral oil.

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3.10 Put-up. Unless otherwise specified, (see 6.2), the rope shall be furnished without knots or splices on non-returnable reels, spools not larger than 6 feet in diameter in the lengths specified in table I. Pieced lengths will be permitted, but no piece shall be less than 600 feet when measured in the relaxed condition. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The reels, spools shall be wound so that each turn and layer is free from entanglement.

3.10.1 When specified, rope shall be put up in coils in one continuous piece without knots or splices in the minimum length specified (see 6.2). A plus tolerance of 10 percent shall be allowed on the length per coil. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The coils shall be wound so that each turn and layer is free from entanglement.

3.11 Moisture content. The moisture content of the rope, as received, shall not exceed 2.0 percent when tested as specified in 4.4.5 (see 6.5).

3.12 Spliceability. The finished rope shall be spliceable and shall not develop yarn displacement or strand cockles in the splicing test specified in 4.4.5.

3.13 Workmanship. The rope shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical characteristics in accordance with 4.4.5.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table II and a contractor's certificate of compliance thereto shall be furnished and will be accepted for the requirements.

TABLE II

Characteristics	Paragraph reference
Material	3.4
Rope size	3.5.2
Single yarn in strand (when applicable)	3.5.2
Plied yarn in strand (when applicable)	3.5.2

4.4.2 End item inspection. The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit shall be one reel, spool, or coil as applicable. Ten percent of the length contained on each sample unit, but not less than 100 feet shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels, spools, or coils each. The acceptable quality level (AQL) shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

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TABLE III. Visual examination defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
	Kink, darting yarn 1/, broken or loose end, bulged strand, strand knot, strand or yarn splice	X	
	Ends not cut off squarely		X
	Ends not securely whipped, taped or heat sealed		X
	Splice or otherwise secured to make a continuous length	X	
Identification	Omitted, incorrect, or illegible		X
	Insecurely attached or not as specified		X
Color	Not as specified	X	
	Resists dyeing or spotty dyeing		X
	Specific shade not within established tolerances		X
	Uneven		X
Cleanness	Spot, or stain, clearly noticeable 2/		X

1/ Darting yarn is an internal yarn which projects through the surface of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.4.3 Examination for length and winding. The sample unit for this examination shall be one reel, spool, or coil. The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 percent defective. The lot size shall be the number of units in the inspection lot. Defects shall be as listed in 4.4.3.1 and 4.4.3.2. The length of the reel, spool, or coil (see table I and 3.10.1) shall be calculated using the following formula:

$$W \times L = \text{Length per holder in feet}$$

Where: W is net weight of the rope in 0.1 pounds

L is the lot average of the weight-length relationship (length pound) as determined in 4.4.5

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4.4.3.1 Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of reel, spool less than specified.

b. Length of coil less than or more than length specified (including permitted tolerances).

c. Length of unit less than marked on ticket.

d. Rope not in continuous length when coils are specified (see 3.10.1).

e. Any piece on reels, spools less than 600 feet in length.

4.4.3.2 Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot or splice or otherwise joining of ends to make a continuous length.

4.4.4 Inspection of preparation for delivery. Packaging, packing and marking shall be inspected in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 defects per 100 units.

4.4.5 Testing of the end item. The methods of testing shall be as specified in FED-STD-191, wherever applicable and as listed in table V. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table IV. The sample unit for testing shall be one reel, spool, or coil of rope. The lot size shall be expressed in units of one reel, spool, or coil. All test reports shall contain the individual values utilized in expressing the final result. The acceptance or rejection of the lot shall be in accordance with table IV. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

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TABLE IV. Samples for tests

Number of coils, reels, spools, or tubes in lot	No. of Samples	Acceptance No. for each test characteristic	Rejection No. for each test characteristic
Up to 25	2	0	1
26 to 150	3	0	1
151 to 1200	4	0	1
1201 and over	7	1	2

TABLE V. Test methods

Characteristic	Requirement reference	Test method
Lay of rope	3.5.1	Visual 2/
Polyester fiber content	3.5.2	4.4.5.1
Number of strands	3.6	Visual 2/
Circumference	Table I	6003
Length per pound	Table I	6004
Breaking force	Table I	6015 3/
Length per 10 turns (30 crowns)	Table I	6000
Elongation	3.6.1	6015 3/
Extractable matter	3.7	2611 4/
Color (shade) matching	3.8	4.4.5.2 2/
Colorfastness to weathering	3.8.1	5671 5/
Identification marker	3.9.1	Visual 1/
Moisture content	3.11	2600
Spliceability	3.12	4.4.5.3 2/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ One determination per sample unit shall be made and the results shall be reported as pass or fail.

3/ Method 6016 of FED-STD-191 may be used to determine breaking force and elongation for ropes 3 inches in circumference or less.

4/ Except that extractable matter shall be determined using the chloroform extraction procedure.

5/ The time of exposure shall be 40 standard fading hours.

4.4.5.1 Polyester fiber content.

4.4.5.1.1 Procedure. Cut each cover yarn specimen to a minimum length of 10 inches and weigh to the nearest milligram. Carefully separate the polyester multifilament yarn component from the polypropylene monofilament core yarn of the cut specimen by unwinding and weigh the polyester portion separately to the nearest milligram.

4.4.5.1.2 Result. The percentage of polyester fiber shall be based on the following equation:

$$\text{Polyester fiber, percent} = \frac{\text{weight of the polyester component}}{\text{total weight of yarn}} \times 100$$

4.4.5.1.3 Five specimens shall be tested from each sample unit, and the polyester fiber component shall be the average of the results obtained from the specimens tested. The results shall be reported to the nearest whole percent.

4.4.5.2 Determination of shade (color) matching. The shade of the dyed rope shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be good approximation to the standard sample under incandescent lamplight at 2800° Kelvin (see 6.3).

4.4.5.3 Determination of spliceability. A breaking strength specimen shall be prepared and spliced at each end with four full tucks. The specimen shall then be allowed to rest for 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

5. PACKAGING

5.1 Preservation - packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The rope, put-up as specified, shall be packaged in accordance with the applicable requirements of MIL-C-3131 (see 6.2).

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C packing. The rope shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. The rope is intended for general purpose use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Circumference (size) required (see table I).
(c) First article. when first article sample is required (see 3.1, 4.3 and 6.4).
(d) Color required, if other than natural (see 3.8).
(e) Put-up and length when other than specified (see 3.10).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) That polyester/polypropylene rope shall be purchased on a price-per-pound basis net weight.
(h) That purchaser will accept at original weight, any unit which has been shortened or cut for test specimens, if in compliance with this specification.

6.3 Standard sample. For access to the standard sample, address the procuring office issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of of one reel, spool, or coil of rope. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

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6.6 Adjustment for dyestuff. A plus tolerance of approximately 10 percent in weight is allowable for dyed rope.

Custodians:

Army - GL
Air Force - 82

Preparing activity:

Army - GL

Project No. 4020-0253

Review activities:

Army - WC
DSA - IS

User activity:

Army - ME

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-43942 ROPE, FIBROUS, POLYESTER/POLYPROPYDENE DUAL FIBER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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